

```
1 import pygame
2 import random
3 x = pygame.init()
4 #colours
5 white = (255,255,255)
6 red = (255,0,0)
7 black =(0,0,0)
8 #creating a window
9 screen_width = 1050
10 screen_height = 500
11 gameWindow = pygame.display.set_mode((screen_width,
    screen_height))
12 pygame.display.set_caption("snake game ")
13 pygame.display.update()
14
15 clock = pygame.time.Clock()
16 font = pygame.font.SysFont(None,50)
17
18 def text_screen(text,colour,x,y):
19     screen_text = font.render(text,True,colour)
20     gameWindow.blit(screen_text,[x,y])
21 def plot_snake(gameWindow,colour,snk_list,snake_size
    ):
22     for x,y in snk_list:
23         pygame.draw.rect(gameWindow,colour,[x,y,
    snake_size,snake_size])
24 def welcome():
25     exit_game = False
26     while not exit_game:
27         gameWindow.fill((233,210,225))
28         text_screen("WELCOME TO THE GAME",black,260,
    250)
29         text_screen("PRESS SPACE BAR TO PLAY",black,
    240,290)
30
31
32     for event in pygame.event.get():
33         if event.type== pygame.QUIT:
34             exit_game = True
35         if event.type == pygame.KEYDOWN:
36             if event.key == pygame.K_SPACE:
```

```
37         gameloop()
38     pygame.display.update()
39     clock.tick(60)
40
41 def gameloop():
42     game_over = False
43     exit_game = False
44     snake_x = 45
45     snake_y = 55
46     velocity_x = 4
47     velocity_y = 4
48     snk_list = []
49     snk_length = 1
50     score = 0
51     init_velocity = 5
52     food_x = random.randint(20, screen_width / 2)
53     food_y = random.randint(20, screen_height / 2)
54
55
56     snake_size = 20
57     fps = 60
58     while not exit_game :
59         if game_over:
60             gameWindow.fill(white)
61             text_screen("GAME OVER! PRESS ENTER TO
CONTINUE",red,100,200)
62             for event in pygame.event.get():
63                 if event.type == pygame.QUIT:
64                     exit_game = True
65                 if event.type == pygame.KEYDOWN:
66                     if event.key == pygame.K_RETURN:
67                         gameloop()
68
69         else:
70
71             for event in pygame.event.get():
72                 if event.type == pygame.QUIT:
73                     exit_game = True
74                 if event.type == pygame.KEYDOWN:
75                     if event.key == pygame.K_RIGHT:
76                         velocity_x = init_velocity
```

```

77         velocity_y = 0
78
79         if event.key == pygame.K_LEFT:
80             velocity_x = -init_velocity
81             velocity_y = 0
82
83         if event.key == pygame.K_UP:
84             velocity_y = -init_velocity
85             velocity_x = 0
86         if event.key == pygame.K_DOWN:
87             velocity_y = init_velocity
88             velocity_x = 0
89         if event.key == pygame.K_q:
90             score+=5
91
92
93     snake_x = snake_x + velocity_x
94     snake_y = snake_y + velocity_y
95     if abs(snake_x-food_x)<8 and abs(snake_y-
food_y)<8:
96         score+= 10
97         print('score:',score*10)
98         food_x = random.randint(20, screen_width / 2
)
99         food_y = random.randint(20, screen_height /
2)
100         snk_length += 5
101
102     gameWindow.fill(white)
103     text_screen('score :'+ str(score),red,5,5)
104
105     pygame.draw.rect(gameWindow,red,[food_x,
food_y,snake_size,snake_size])
106     head = []
107     head.append(snake_x)
108     head.append(snake_y)
109     snk_list.append(head)
110     if len(snk_list)>snk_length:
111         del snk_list[0]
112     if head in snk_list[:-1]:
113         game_over = True

```

```
114         if snake_x<0 or snake_x> screen_width or
            snake_y<0 or snake_y>screen_height:
115             game_over = True
116             print("Game over")
117             plot_snake(gameWindow,black,snk_list,
            snake_size)
118             pygame.display.update()
119             clock.tick(fps)
120             pygame.quit()
121             quit()
122 welcome()
123
```